

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Fourth Semester of M. Sc. (Mathematics) Examination April-May 2019
MA824 : Special Functions

Date : 01-05-2019, Wednesday Time : 1:30 pm to 04:30 pm Total marks 70

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Maximum time to answer Multiple Choice Questions will be 45 minutes.
4. Each question carries one mark.
5. Notations and terminologies are standard.

Q-I Choose the correct answer from the given options in the following:

[20]

(1) The infinite product $\prod_{n=2}^{\infty} \left(1 - \frac{1}{n^2}\right)$ converges to _____.

- | | |
|-------|---------|
| (a) 0 | (b) 0.5 |
| (c) 1 | (d) 1.5 |

(2) If $\prod_{n=1}^{\infty} (1 + a_n)$, $a_n \neq -1$ converges, then $\lim_{n \rightarrow \infty} a_n =$ _____.

- | | |
|-------|-------|
| (a) 0 | (b) 2 |
| (c) 4 | (d) 6 |

- (3) The approximate value of the Euler constant ' γ ' is _____.
 (a) 0.3742 (b) 0.4752
 (c) 0.5772 (d) 0.6772
- (4) Simple pole of $\Gamma(3+z)$ is $z =$ _____.
 (a) -3 (b) -2
 (c) -1 (d) 0
- (5) If _____, then $\int_0^{\infty} e^{-t} t^{z-1} dt$ exists.
 (a) $-2 < \operatorname{Re}(z)$ (b) $-2 < \operatorname{Re}(z) < -1$
 (c) $-1 < \operatorname{Re}(z) < 0$ (d) $\operatorname{Re}(z) > 0$
- (6) If $k =$ _____, then $(-5)_k$ is a nonzero constant.
 (a) 5 (b) 6
 (c) 7 (d) 8
- (7) The value of $\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)$ is _____.
 (a) π (b) $2\sqrt{\pi}$
 (c) $\pi\sqrt{2}$ (d) 2π
- (8) A sufficient condition for absolute convergence of ${}_2F_1[a, b; c; 1]$ is _____.
 (a) $\operatorname{Re}(c) < 0$ (b) $\operatorname{Re}(c-a) > 0$
 (c) $\operatorname{Re}(c-b) > 0$ (d) $\operatorname{Re}(c-a-b) > 0$
- (9) The hypergeometric function ${}_2F_1[1, 3; 5; 1]$ converges to _____.
 (a) 2 (b) 4
 (c) 6 (d) 8
- (10) Total number of contiguous function relations of ${}_2F_1[a, b; c; z]$ is _____.
 (a) 11 (b) 13
 (c) 15 (d) 17
- (11) If _____, then the series ${}_pF_q[z]$ converges for all finite z .
 (a) $p > q$ (b) $pq < 1$
 (c) $pq = q+1$ (d) $p \leq q$
- (12) The order of differential equation satisfied by ${}_3F_7[z]$ is _____.
 (a) 7 (b) 8
 (c) 9 (d) 10

- (13) The hypergeometric function ${}_3F_2 \left[\begin{matrix} a, & b, & c; \\ 3+a-\alpha-b, & 1+a-c; \end{matrix} x \right]$ is said to be well poised, if $\alpha =$ _____.
 (a) 1 (b) 2
 (c) 3 (d) 4
- (14) If $\alpha =$ _____, then the function ${}_3F_2 \left[\begin{matrix} -n, & a, & b; \\ c, & 2-\alpha-c+a+b-n; \end{matrix} 1 \right]$ is Saalschutzhian.
 (a) 1 (b) 2
 (c) 3 (d) 4
- (15) The interval of convergence of ${}_3F_2 \left[\begin{matrix} 5, & 8, & 9; \\ 8, & 9; \end{matrix} \frac{x}{2} \right]$ is _____.
 (a) $(-2, 2)$ (b) $(-3, 3)$
 (c) $(-4, 4)$ (d) $(-5, 5)$
- (16) The order of Bessel's differential equation is _____.
 (a) 1 (b) 2
 (c) 3 (d) 4
- (17) The differential recurrence relation satisfied by Bessel function is _____.
 (a) $\frac{d}{dz} [z^n J_n(z)] = -z^n J_{n-1}(z)$ (b) $\frac{d}{dz} [z^{-n} J_n(z)] = -z^{-n} J_{n+1}(z)$
 (c) $\frac{d}{dz} [z^{-n} J_n(z)] = z^n J_{n+1}(z)$ (d) $\frac{d}{dz} [z^n J_n(z)] = J_{n-1}(z)$
- (18) The Bessel function $J_{\frac{1}{2}} \left(\frac{\pi}{2} \right)$ converges to _____.
 (a) 1 (b) 2
 (c) $\frac{\pi}{2}$ (d) $\frac{2}{\pi}$
- (19) The hypergeometric function representation of exponential function e^x is _____.
 (a) ${}_0F_0[-; -; x]$ (b) ${}_1F_0[1; -; x]$
 (c) ${}_0F_1[-; 1; x]$ (d) ${}_2F_1[1, 2; 1; x]$
- (20) The Generating function relation of the Hermite polynomial is given by

$$\sum_{n=0}^{\infty} \frac{H_n(x) t^n}{n!} = \text{_____}.$$

 (a) $e^{(2xt-t^2)}$ (b) $e^{(1-2xt+t^2)}$
 (c) $(1-2xt+t^2)$ (d) $\sqrt{1-2xt+t^2}$

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Instructions:

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3. Figures to the right indicate marks.
4. Notations and terminologies are standard.

SECTOIN - I

Q-II Answer the following:

(A) Attempt any TWO of the following: [8]

- (1) State and prove Legendre duplication formula.
- (2) State and prove simple transformation of ${}_2F_1[z]$.
- (3) State and prove Saalschutz' theorem.

(B) Attempt any SIX of the following: [12]

- (1) State relation between Beta function and Gamma function and show that $\beta(p, q) = \beta(p+1, q) + \beta(p, q+1)$.

- (2) Evaluate $\int_{-\infty}^{\infty} e^{-x^2} dx$.

- (3) Show that $(1-x)^{-a} = \sum_{n=0}^{\infty} \frac{(a)_n}{n!} x^n$ for $a \in \mathbb{C}$. Also show that it converges for $|x| < 1$.

- (4) Show that ${}_2F_1[-n, b; c; 1] = \frac{(c-b)_n}{(c)_n}$.

- (5) Find $\frac{d}{dz} {}_pF_q \left[\begin{matrix} a_1, \dots, a_p; \\ b_1, \dots, b_q; \end{matrix} z \right]$.

- (6) State and prove simple integral representation of ${}_pF_q[z]$.

- (7) Show that ${}_{(q+1)}F_q[z]$ converges for $|z| < 1$.

- (8) If b is neither zero nor a negative integer, then show that

$${}_1F_1[a; b; z] = e^z {}_1F_1[b-a; b; -z].$$

- (9) Show that $e^t {}_0F_1[-; 1+\alpha; -xt] = \sum_{n=0}^{\infty} \frac{L_n^{(\alpha)}(x)}{(1+\alpha)_n} t^n$.

SECTOIN - II

Q-III Answer the following:

(A) Attempt any THREE of the following: [18]

- (1) If z is non-integral, then show that $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$.
- (2) Define Contiguous function and contiguous function relation. Show that $[a + (b-c)z]F = a(1-z)F(a+) - c^{-1}(c-a)(c-b)zF(c+)$.
- (3) State and prove Whipple's theorem for terminating series.
- (4) Derive value of $J_{\frac{1}{2}}(z)$ and $J_{-\frac{1}{2}}(z)$. Using it find $J_{\frac{3}{2}}(z)$.
- (5) Show that ${}_1F_1[a; 2a; z] = e^z {}_1F_1[a; 2a; -z]$.

(B) Attempt any THREE of the following: [12]

- (1) Show that if z is not a negative integer, then show that $\lim_{n \rightarrow \infty} \frac{(n-1)!n^z}{(z)_n}$ exists.
- (2) If $2b$ is neither zero nor a negative integer and if $|y(1-y)^{-1}| < 1$ and $|2y| < 1$, then show that
$$(1-y)^{-a} {}_2F_1\left[\frac{a}{2}, \frac{a+1}{2}; \frac{2b+1}{2}; \frac{y^2}{(1-y)^2}\right] = {}_2F_1[a, b; 2b; 2y].$$
- (3) Prove that if $g_n = {}_2F_1[-n, \alpha; 1+\alpha-n; 1]$ and α is not an integer, then $g_n = 0$ for $n \geq 1$ and $g_0 = 1$.
- (4) If $K(x) = \frac{\pi}{2} {}_2F_1\left[\frac{1}{2}, \frac{1}{2}; 1; x^2\right]$, then show that

$$\int_0^t K(\sqrt{x(t-x)}) dx = \pi \operatorname{Arcsin}\left(\frac{t}{2}\right).$$

- (5) Show that $J_n(z) = \frac{z^n}{\sqrt{\pi}\Gamma\left(n + \frac{1}{2}\right)2^{n-1}} \int_0^{\frac{\pi}{2}} \sin^{2n} \alpha \cos(z \cos \alpha) d\alpha$.
